



THE LONDON SCHOOL
OF ECONOMICS AND
POLITICAL SCIENCE ■

Summer 2023 Exam

MY457

Causal Inference for Observational and Experimental Studies

Suitable for all candidates

Instructions to candidates

This paper contains four questions. Answer all four questions.

Question 1 is worth 40 marks in total, Question 2 20 marks, Question 3 20 marks, and Question 4 20 marks. The marks in brackets indicate marks available for each part of each question.

Time allowed **Reading Time:** *None*
 Writing Time: *2 hours*

You are supplied with: *No additional materials*

You may also use: *No additional materials*

Calculators: *Calculators are allowed in this exam*

1. (a) Explain the notation Y_{id} assuming that $D \in \{0, 1\}$ represents a binary treatment for a study in which we observe individuals $i = \{1, \dots, N\}$. [2 marks]
- (b) Given the above definitions and explanation, explain the notation $Y_{i1}|D_i = 1$. [2 marks]
- (c) Contrast the meaning of Y_{i0} with the meaning of $Y_{i0}|D_i = 0$. [2 marks]
- (d) Contrast the meaning of $Y_{i0}|D_i = 1$ with the meaning of $Y_{i0}|D_i = 0$. [2 marks]
- (e) When would $Y_{i0} = Y_i$, where Y_i is the observed outcome for unit i ? [2 marks]
- (f) The table below presents a hypothetical study involving twelve units. You can assume these units represent the entire population (i.e., set aside issues related to sampling uncertainty). Potential outcomes are given as Y_1 and Y_0 for treatment $D \in \{0, 1\}$, where 1 represents treatment and 0 control, and $X \in \{0, 1\}$ is a binary pre-treatment covariate.

Unit	Y_0	Y_1	D	X
1	28	28	0	0
2	16	22	0	0
3	24	26	1	0
4	22	22	1	0
5	5	4	1	0
6	12	14	0	0
7	25	29	0	1
8	20	24	1	1
9	14	20	0	1
10	16	21	0	1
11	28	30	1	1
12	13	17	1	1

- i. Calculate **both** the true average treatment effect (ATE) and the true average treatment effect on the treated (ATT). Be sure to show your working. [6 marks]
- ii. How many of the 24 potential outcomes in the table will be observed? [2 marks]
- iii. Estimate the ATE and ATT using only the observed outcomes. Be sure to show your working. [2 marks]

(g) Consider again the data introduced in 1.f.

- i. Calculate the propensity score for being assigned to the treatment group for each value of $X \in \{0, 1\}$. [2 marks]
- ii. Using the values in the table, calculate the true conditional average treatment effect (CATE) when $X = 0$. [2 marks]
- iii. Calculate the true CATE when $X = 1$. [2 marks]
- iv. Considering the answers you have given for (1.g.i) - (1.g.iii), what might you conclude about the role of X in this particular study? Pay attention to both the assignment mechanism and to treatment effect moderation. [4 marks]

(h) Consider a case where Y_0 and Y_1 are the potential outcomes for Y given a treatment indicator $D \in \{0, 1\}$. We also observe a binary pre-treatment covariate $X \in \{0, 1\}$. In the notation used below:

- $E(A|B = b)$ denotes a conditional expected value of the random variable A for all units in the population where $B = b$.
- $p(B = b)$ denotes proportions of the different values of B in the population.

Consider the following derivation:

$$\sum_{x=0}^1 [E(Y|D = 1, X = x) - E(Y|D = 0, X = x)] \cdot p(X = x) \quad (1)$$

$$= \sum_{x=0}^1 [E(Y_1|D = 1, X = x) - E(Y_0|D = 0, X = x)] \cdot p(X = x) \quad (2)$$

$$= \sum_{x=0}^1 [E(Y_1|X = x) - E(Y_0|X = x)] \cdot p(X = x) \quad (3)$$

$$= E(Y_1) - E(Y_0) \quad (4)$$

$$= ATT \quad (5)$$

- i. Line (1) in the above derivation represents an estimator. What is the name of that estimator? [2 marks]
- ii. Explain, in your own words, what this derivation is intended to show. [3 marks]
- iii. The derivation relies on an assumption that permits us to move from line (2) to line (3). Name that assumption and then, in your own words, explain what it means. [5 marks]

2. The table below presents a hypothetical study involving twelve units. Potential outcomes are given as Y_1 and Y_0 for treatment $D \in \{0, 1\}$, where 1 represents treatment and 0 control, and X is a continuous pre-treatment covariate.

Unit	Y_0	Y_1	D	X
1	5.3	8.6	0	1.4
2	5.4	9.7	1	2.4
3	4.9	9.3	1	2.7
4	4.8	10.5	0	3.1
5	5.3	10.0	1	3.4
6	4.1	10.3	1	3.5
7	5.3	11.2	0	3.6
8	4.9	10.7	1	3.6
9	4.9	11.2	0	3.8
10	4.5	11.5	0	3.8
11	4.9	15.2	1	8.5
12	5.9	15.5	0	9.3

- (a) Let us consider a matching estimator for the average treatment effect on the treated (ATT), matching on their observed values of X .

- i. The following table displays these values such that the values of X have been arrayed in ascending order within the control group and within the treatment group. Assuming 1 : 1 **nearest-neighbour matching with replacement**, which treated values of X are matched with which control values of X ? Copy this table into your answer booklet, then draw a line between each pair of values of X that are matched to answer the question. [3 marks]

$D = 0$	$D = 1$
1.4	2.4
3.1	2.7
3.6	3.4
3.8	3.5
3.8	3.6
9.3	8.5

- ii. Now conduct matching as described above for the units in the study. Copy the following table into your answer booklet, and fill in these columns, assuming that $\hat{\tau}_i$ is the matching estimate of the individual treatment effect τ_i . [6 marks]

Treated Unit	$Y_{treated}$	$X_{treated}$	Matched Unit	$Y_{matched}$	$X_{matched}$	$\hat{\tau}_i$
2						
3						
5						
6						
8						
11						

- iii. Estimate the ATT from your table. Be sure to show your working. [4 marks]

- (b) Assuming that the Stable Unit Treatment Value Assumption (SUTVA) holds, and that conditional ignorability also holds, name and explain the additional assumption required for the matching estimator to identify the ATT? How would this assumptions change if we wished to identify the average treatment effect (ATE) instead? [5 marks]
- (c) Now consider regression-based estimator for the average treatment effect (ATE). We fit the following linear regression:

$$Y_i = \hat{\alpha} + \hat{\tau}D_i + \hat{\beta}X_i + \hat{\epsilon}_i$$

where $i = 1, \dots, n = 12$ denote the units, and $\hat{\epsilon}_i$ the estimated residuals, and $\hat{\tau}$ is the estimator for the ATE.

What does the linear regression specified above assume about the ATE? [2 marks]

3. (a) Briefly describe in your own words the basic idea, key elements, and identification assumptions of the **instrumental variables design**. Use notation as appropriate. Explain what causal effects could potentially be estimated under this design, and how they might be estimated. When is this design most defensible? What are some of its weaknesses? [10 marks]
- (b) Briefly describe in your own words the basic idea, key elements, and identification assumptions of the **sharp regression discontinuity design**. Use notation as appropriate, and include a graph if helpful. Explain what causal effects could potentially be estimated under this design, and how they might be estimated. When is this design most defensible? What are some of its weaknesses? [10 marks]

4. In this question, we will imagine that Britain, taking inspiration from the United States, decides to experiment with policy federalism. The government mandates that a number of non-randomly selected British counties are required to raise the Living Wage (read: minimum wage), while other counties will be exempt from this change. Let us call this treatment $D \in \{0, 1\}$, where 1 represents being exposed to the change, and 0 represents being exempt.

In the table below we have measured average full time employment rates for each set of counties (those that experience the policy change, and those that do not), before and after the change was implemented. Let us call this our observed outcome Y .

Policy Status	Pre-treatment	Post-treatment
Living Wage Policy Changed	5.5%	5.75%
No Policy Change	8%	11%

- (a) What research design might be appropriate in this setting? [2 marks]
- (b) Estimate the ATT for the post-period in percentage point terms. [3 marks]
- (c) Assume that SUTVA holds in this case. Given this:
- Name and explain the additional critical assumption that underpins identification in this design [4 marks]
 - In general, can one test this assumption directly? Why or why not? [2 marks]
 - If you could go out and collect more data, how might you probe the assumption empirically? [4 marks]
- (d) Draw a graph that plots the data in the table using two solid lines (one for the treated group and one for the control group). Add a dotted line to plot the missing potential outcome, assuming the assumption you outlined above holds. [5 marks]

END